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LSP

Task - 6: Privilege levels in linux - Kernel mode and user mode

1> Kernel Mode - is the most privileged level. It provides the core part of the operating system with unfettered access to every system resource, including hardware and memory.

Characteristics

- Complete access to every CPU instruction and hardware component.
- Direct communication with device drivers, memory management and I/O operations.
- Kernel code executes in this mode, which means one mistake will crash the whole system.

Examples of operations in Kernel mode.

1. Device driver communication - A Kernel mode printer driver is able to directly communicate with printer hardware.
2. Process scheduling - The kernel chooses the next process to execute according to scheduling algorithm.
3. Handling system calls - If a user-mode program wants a service, the kernel manages it.

2> User mode - The lower privilege level, user mode contains applications and processes that run on it without any direct access to hardware or kernel memory.

Characteristics

- i) Restricted access to system resources.
- ii) Cannot run privileged CPU instructions.
- iii) If a program crashes, it does not crash the entire system - just that process.

Examples of operations in user mode.

- i) Running applications - web browsers, word processors and games all run in user mode.
- ii) Standard library calls - calls such as 'printf()' or 'malloc()' run in user mode and only access system resources via system calls.
- iii) Application logic - Processing of user input, UI updates and calculations happen in this mode.

3) Transition Between modes.

The switch from user mode to kernel mode occurs via system calls. A system call is a governed interface by which user programs are able to ask the kernel for services. This guarantees that although the kernel executes imperative operations, they're accessed in a secure, controlled way.

Example -

```
int fd = open("example.txt", O_RDONLY);
```